BOOK RECOMMENDATION SYSTEM

1. ABSTRACT

This project focuses on the development of a Book Recommendation System using Python. The system suggests books to users based on their preferences, past interactions, and similarity with other users. It uses a rule-based collaborative filtering approach instead of machine learning, making it simple and efficient.

The system provides a command-line interface where users can register, login, search books, rate books, and receive personalized recommendations. It also includes mood-based recommendations and popularity-based suggestions for new users.

All data is stored in CSV files, ensuring lightweight and easy data handling. The project demonstrates how recommendation systems can be built using core programming concepts without complex algorithms.

2. INTRODUCTION

In today's digital world, users are overwhelmed with a large number of book choices. Finding the right book based on personal interest can be difficult. Recommendation systems help solve this problem by suggesting relevant items based on user behavior.

This project implements a Book Recommendation System that allows users to:

- Register and login
- Search for books
- Rate books
- Get personalized recommendations
- Get mood-based suggestions

The system is developed using Python and uses CSV files instead of databases, making it simple and beginner-friendly. It follows a modular structure where each functionality is handled separately.

☞ A structured report typically includes abstract, introduction, methodology, and conclusion to clearly explain the system and its outcomes.

3. OBJECTIVES

- To develop a book recommendation system using Python
- To implement user authentication (login/register)
- To recommend books based on user similarity
- To provide mood-based recommendations
- To allow users to search and rate books
- To manage data using CSV files
- To build a simple CLI-based system

4. SYSTEM MODULES

4.1 User Authentication Module

- Register new users
- Login existing users
- Store user credentials in CSV

4.2 Book Management Module

- Load books from CSV
- Store book ID, title, and genre
- Allow admin to add books

4.3 Recommendation Module

- Uses similarity between users
- Suggests books liked by similar users
- Uses cosine similarity logic

4.4 Mood-Based Recommendation Module

- Suggests books based on mood:
  - Happy → Comedy, Romance
  - Sad → Drama
  - Motivated → Self-help, Biography

4.5 Search Module

- Search books using keywords
- Displays matching results

4.6 Rating Module

- Users can rate books (1–5)
- Stores ratings in interactions file

4.7 Popularity Module

- Recommends most popular books
- Useful for new users

4.8 Error Handling & Logging Module

- Handles invalid inputs
- Logs errors into file
- Prevents system crash

5. FEATURES OF THE SYSTEM

- User login and registration
- Personalized recommendations
- Mood-based suggestions
- Search functionality
- Book rating system
- CSV-based data storage
- Error handling and logging
- Simple command-line interface

6. USAGE OF THE SYSTEM

Steps to use the system:

1. User registers or logs in
2. User can search books
3. User rates books
4. System generates recommendations
5. User can select mood for suggestions
6. Popular books shown for new users

The system ensures smooth and efficient recommendations using simple logic.

7. DATA STORAGE (CSV STRUCTURE)

📁 books.csv

```
book_id,title,genre
1,The Alchemist,Fiction
2,Atomic Habits,Self-help
```

📁 users.csv

```
username,password
shivani,1234
```

📁 interactions.csv

```
user_id,book_id,rating
shivani,1,5
```

8. RECOMMENDATION LOGIC

The system uses **collaborative filtering logic**:

- Finds similar users based on ratings
- Calculates similarity using cosine formula

- Recommends books liked by similar users

Logic:

- Similar users → similar preferences
- Weighted ratings → recommendation score

9. ADVANTAGES

- Simple and easy to use
- No machine learning required
- Lightweight (CSV-based)
- Easy to understand logic
- Beginner-friendly project
- Error-resistant system

10. LIMITATIONS

- Command-line interface only
- No real-time updates
- Limited dataset scalability
- No advanced AI/ML
- Basic recommendation logic

11. FUTURE ENHANCEMENTS

- Add graphical user interface (GUI)
- Develop web application (Flask/Django)
- Add machine learning algorithms
- Improve recommendation accuracy
- Add user profiles and history
- Integrate APIs (Google Books, etc.)

12. CONCLUSION

The Book Recommendation System successfully demonstrates how personalized recommendations can be generated using simple Python logic without machine learning.

The project highlights key programming concepts such as:

- File handling (CSV)
- Data structures
- Algorithm design
- Modular programming
- Error handling

With further enhancements like UI and AI integration, this system can evolve into a full-scale recommendation platform.